QUERY PROCESSING FOR DATA SCIENCE

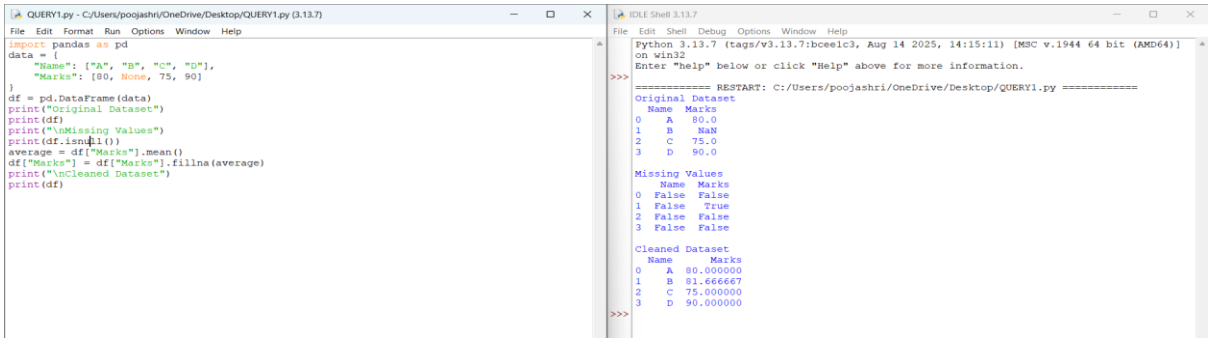
Question 2: Missing Data Handling
Problem Explanation

In real-world datasets, some values may be missing due to incomplete data collection or human error. Missing values can affect analysis and machine learning models. This program identifies missing values, replaces them with the average of the respective column, and displays the cleaned dataset.

CODE

```
import pandas as pd
data = {
    "Name": ["A", "B", "C", "D"],
    "Marks": [80, None, 75, 90]
}
df = pd.DataFrame(data)
print("Original Dataset")
print(df)
print("\nMissing Values")
print(df.isnull())
average = df["Marks"].mean()
df["Marks"] = df["Marks"].fillna(average)
print("\nCleaned Dataset")
print(df)
```

OUTPUT



CONCLUSION

The program successfully detects missing values and replaces them with the column average, resulting in a complete and cleaner dataset for analysis.

Question 4: Sorting Dataset

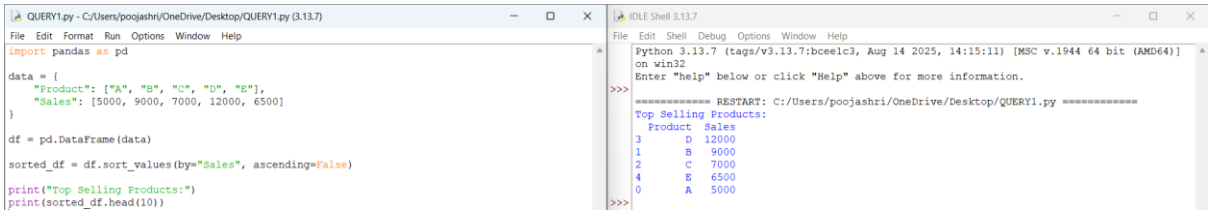
Problem Explanation

Sorting data helps identify the highest and lowest values quickly. This program sorts products based on sales amount and displays the top-selling products.

CODE

```
import pandas as pd

data = {

    "Product": ["A", "B", "C", "D", "E"],

    "Sales": [5000, 9000, 7000, 12000, 6500]

}

df = pd.DataFrame(data)

sorted_df = df.sort_values(by="Sales", ascending=False)

print("Top Selling Products:")

print(sorted_df.head(10))
```

OUTPUT



CONCLUSION

The program arranges products in descending order of sales, making it easy to identify the best-performing products.

Question 6: Data Cleaning

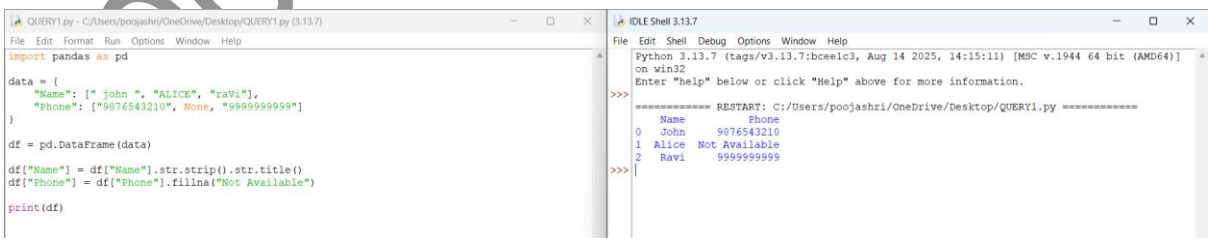
Problem Explanation

Datasets often contain extra spaces, inconsistent letter cases, and missing values. Cleaning the data improves its quality and ensures accurate analysis.

Python Code

```
import pandas as pd

data = {
    "Name": [" john ", "ALICE", "raVi"],
    "Phone": ["9876543210", None, "9999999999"]
}

df = pd.DataFrame(data)

df["Name"] = df["Name"].str.strip().str.title()

df["Phone"] = df["Phone"].fillna("Not Available")

print(df)
```

OUTPUT



Conclusion

The dataset becomes cleaner by removing unnecessary spaces, formatting names properly, and filling missing phone numbers.

Question 7: Data Transformation

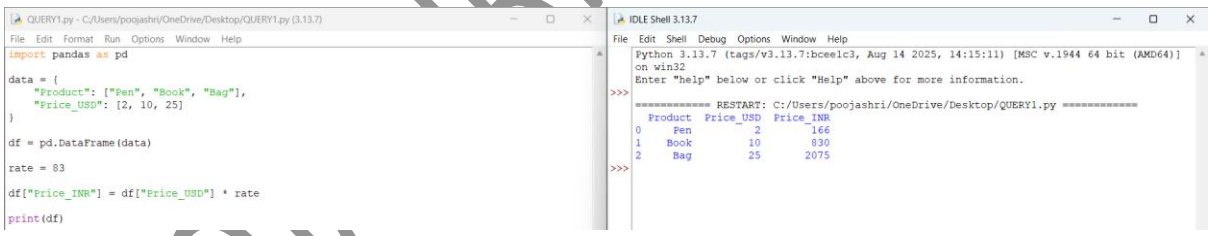
Problem Explanation

Data transformation converts data into another format for better usability. This program converts product prices from USD to INR by creating a new column.

Python Code

```
import pandas as pd

data = {
    "Product": ["Pen", "Book", "Bag"],
    "Price_USD": [2, 10, 25]
}

df = pd.DataFrame(data)

rate = 83

df["Price_INR"] = df["Price_USD"] * rate

print(df)
```

OUTPUT



Conclusion

The program converts prices into Indian Rupees and stores them in a new column for easier understanding.

Question 10: Merge Two CSV Files

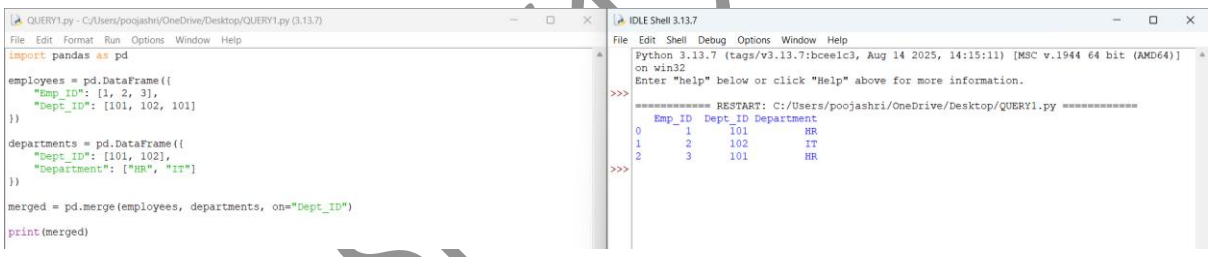
Problem Explanation

Sometimes information is stored in multiple files. Merging datasets combines related information using a common key such as Department ID.

Python Code

```
import pandas as pd
employees = pd.DataFrame({
    "Emp_ID": [1, 2, 3],
    "Dept_ID": [101, 102, 101]
})
departments = pd.DataFrame({
    "Dept_ID": [101, 102],
    "Department": ["HR", "IT"]
})
merged = pd.merge(employees, departments, on="Dept_ID")
print(merged)
```

OUTPUT



Conclusion

The program combines employee and department information into one dataset using the common Department ID.

Question 12: Convert CSV into JSON

Problem Explanation

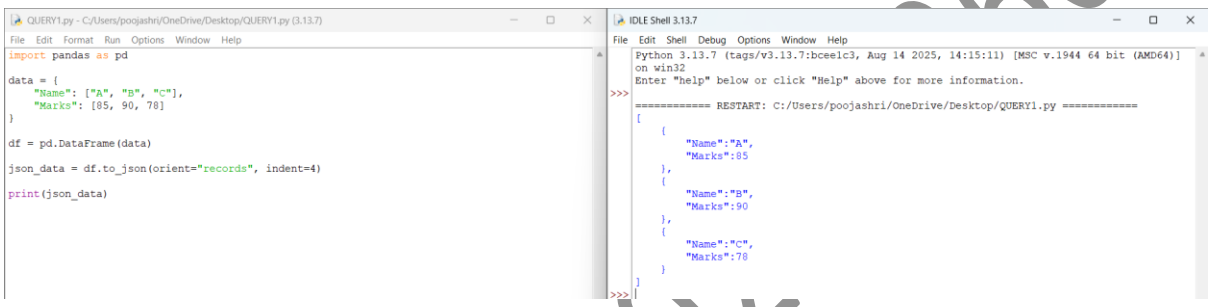
JSON is a popular data format used in web applications. This program converts CSV data into JSON format for easier data exchange.

Python Code

```
import pandas as pd
```

```
data = {
    "Name": ["A", "B", "C"],
    "Marks": [85, 90, 78]
}
df = pd.DataFrame(data)
json_data = df.to_json(orient="records", indent=4)
print(json_data)
```

OUTPUT



Conclusion

The program extracts only the required employee records using filtering conditions.

Question 13: JSON Data Filtering

Problem Explanation

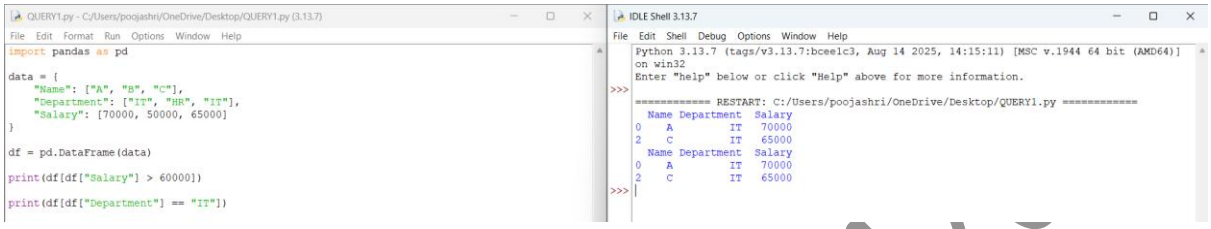
Filtering allows us to display only the required records from a dataset. This program filters employees with salary greater than ₹60,000 and those working in the IT department.

Python Code

```
import pandas as pd
data = {
    "Name": ["A", "B", "C"],
    "Department": ["IT", "HR", "IT"],
    "Salary": [70000, 50000, 65000]
```

```
df = pd.DataFrame(data)

print(df[df["Salary"] > 60000])

print(df[df["Department"] == "IT"])
```

OUTPUT



Conclusion

The program extracts only the required employee records using filtering conditions.

Question 16: XML Data Analysis

Problem Explanation

XML files store structured information. This program reads XML data and displays the total number of books, books published after 2020, and books costing more than ₹500.

Python Code

```
import pandas as pd


df = pd.DataFrame({
    "Book": ["A", "B", "C"],
    "Year": [2019, 2022, 2021],
    "Price": [450, 600, 750]
})


print("Total Books:", len(df))
```

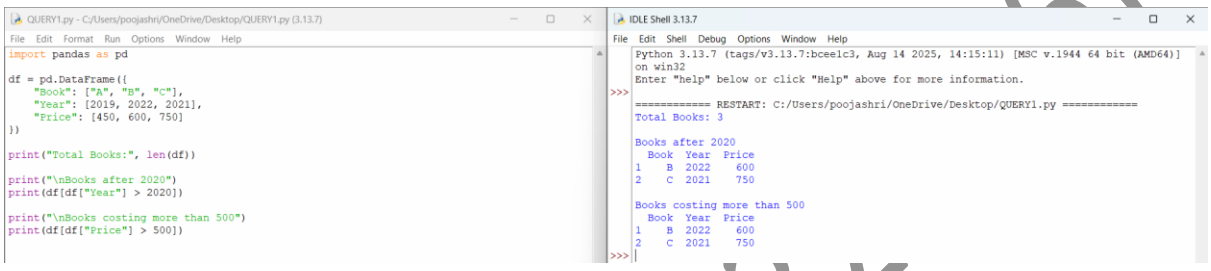
print("\nBooks after 2020")

print(df[df["Year"] > 2020])

print("\nBooks costing more than 500")

print(df[df["Price"] > 500])

OUTPUT



Conclusion

The program analyzes book information by counting records and applying filtering conditions.

Question 18: Dictionary-Based Analysis

Problem Explanation

Python dictionaries store data in key-value pairs. This program stores employee salaries, identifies the highest and lowest salary, and calculates the average salary.

Python Code

```
employees = {
    "Rahul": 50000,
    "Anita": 70000,
    "Vijay": 60000
}

highest = max(employees, key=employees.get)
```

lowest = min(employees, key=employees.get)

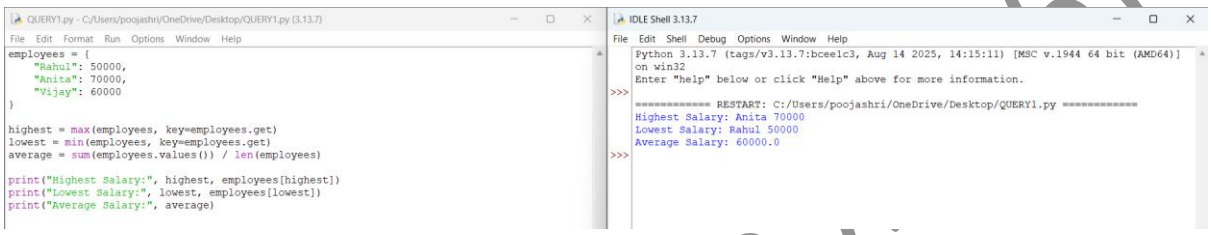
average = sum(employees.values()) / len(employees)

print("Highest Salary:", highest, employees[highest])

print("Lowest Salary:", lowest, employees[lowest])

print("Average Salary:", average)

OUTPUT



Conclusion

The program uses dictionaries to efficiently store employee information and perform salary analysis such as finding the highest, lowest, and average salary.